

R-Universe RFG-R

A regular multiscale completion with a scalar-action audit and empirical protocol

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Abstract. *RFG-R is a precisely specified low-energy completion of the R-Universe preferred-foliation branch. It regularizes the $X=0$ weak-field singularity while retaining the reconstructed cosmological background and luminal tensor propagation to $O((\epsilon/X)^p)$. The completion has two explicit domains: a cosmological relational action and a locally GR-matched weak-field EFT. The potential is reconstructed by quadrature from the prescribed branch rather than postulated. The local domain has $\gamma=\beta=1$ and $\alpha_1=\alpha_2=0$. The exact ADM-to-extended-EFT map contains a nonzero $\bar{m}_5 \delta R^3 \delta K$ operator. An executed pure-gravity scalar audit retains that operator and finds a degenerate $\zeta\text{-dot-squared}$ coefficient on the reference branch. Thus a complete sourced photon--baryon--CDM--neutrino reduction is required before spectra or a CMB likelihood can be evaluated. This paper defines the model, the audited limitation, and the empirical protocol; it makes no data-preference claim.*

Keywords: modified gravity; preferred foliation; CMB; post-Newtonian tests; effective field theory; standard sirens

1. The completion problem

The R-Universe homogeneous branch is predictive for $H(a)$, the effective relational equation of state, and the tensor distance ratio. Its original response $Q(X)=1-A X^{-(\theta)}$, however, is singular in the $X=0$ limit used by a standard static weak-field expansion. A genuine empirical theory therefore needs a definition of its local limit as well as a derived linear matter and radiation system. RFG-R supplies the local completion and the action-level inputs for the latter. It is deliberately an effective low-energy model: cosmological geometry activates the relational response, while resolved local gravity is matched to General Relativity.

2. Regular cosmological response

Let p be even and larger than θ , with reference values $\theta=1.6$, $p=4$, and $\epsilon=10^{-8}$. Define $\nu=1+\theta/p$ and $A=\Omega_{R0}/(1+\theta)$.

$$\begin{aligned} R_{\epsilon}(X) &= \Omega_{R0} X^{(p+2)}/(X^p+\epsilon^p)^{\nu} \\ Q_{\epsilon}(X) &= 1 - A X^p/(X^p+\epsilon^p)^{\nu} \end{aligned}$$

The covariant preferred-foliation action is the original RFG action after replacing Q and V by Q_{ϵ} and V_{ϵ} . Matter remains minimally and universally coupled to the Jordan metric.

$$\begin{aligned} S_{\text{cos}} &= (M_{\text{Pl}}^2/2) \int d^4x \sqrt{-g} \\ &\quad \{ Q_{\epsilon}(X)[R^3 + K_{mn} K^{mn} - K^2] + 2 H_0^2 V_{\epsilon}(X) \} \\ &\quad + S_m[g, \Psi] \end{aligned}$$

The FLRW lapse equation is fixed to be

$$E^2(a) = \Omega_{m0} a^{-3} + \Omega_{r0} a^{-4} + R_{\epsilon}(E)$$

and determines the potential without a new function:

$$\begin{aligned} F_{\epsilon} &= X^2 - R_{\epsilon} - X^2 Q_{\epsilon} - X^3 dQ_{\epsilon}/dX \\ V_{\epsilon}(X) &= -3 X \int_0^X ds F_{\epsilon}(s)/s^2 \end{aligned}$$

The integral is finite because F_{epsilon}/s^2 is $O(X^p)$ at the origin. A term proportional to X is an ADM boundary representative and has no field-equation content.

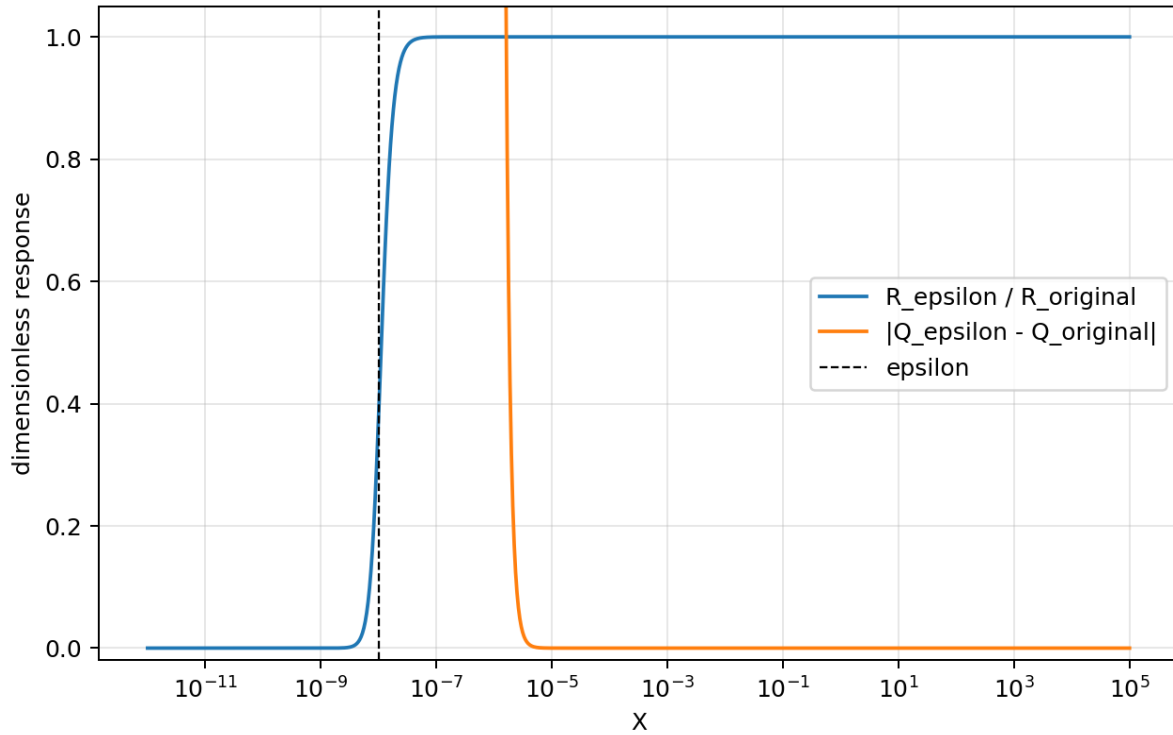


Figure 1. The regularized response becomes the original response immediately above the epsilon transition. On the cosmological branch the analytic difference is $O((\text{epsilon}/X)^4)$.

3. Exact limits

At X much larger than epsilon the regularized branch recovers the original one. With the reference epsilon, the analytic relative correction is below $2.5e-32$ for $X \geq 0.8$; double precision evaluates the equality at about $1e-15$. At $X=0$ the action is regular.

$$\begin{aligned} Q_{\text{epsilon}} &= 1 - A \text{epsilon}^{-(p+\theta)} X^p + O(X^{2p}) \\ V_{\text{epsilon}} &= -[3 \Omega_{R0}(p-\theta)/((1+\theta)(p+1)\text{epsilon}^{(p+\theta)})] X^{(p+2)} \\ &\quad + O(X^{(2p+2)}) \end{aligned}$$

For $p=4$ there is no physical correction to the quadratic Einstein-Hilbert ADM operator. This removes the $X=0$ singularity without altering the background or tensor branch that motivated RFG.

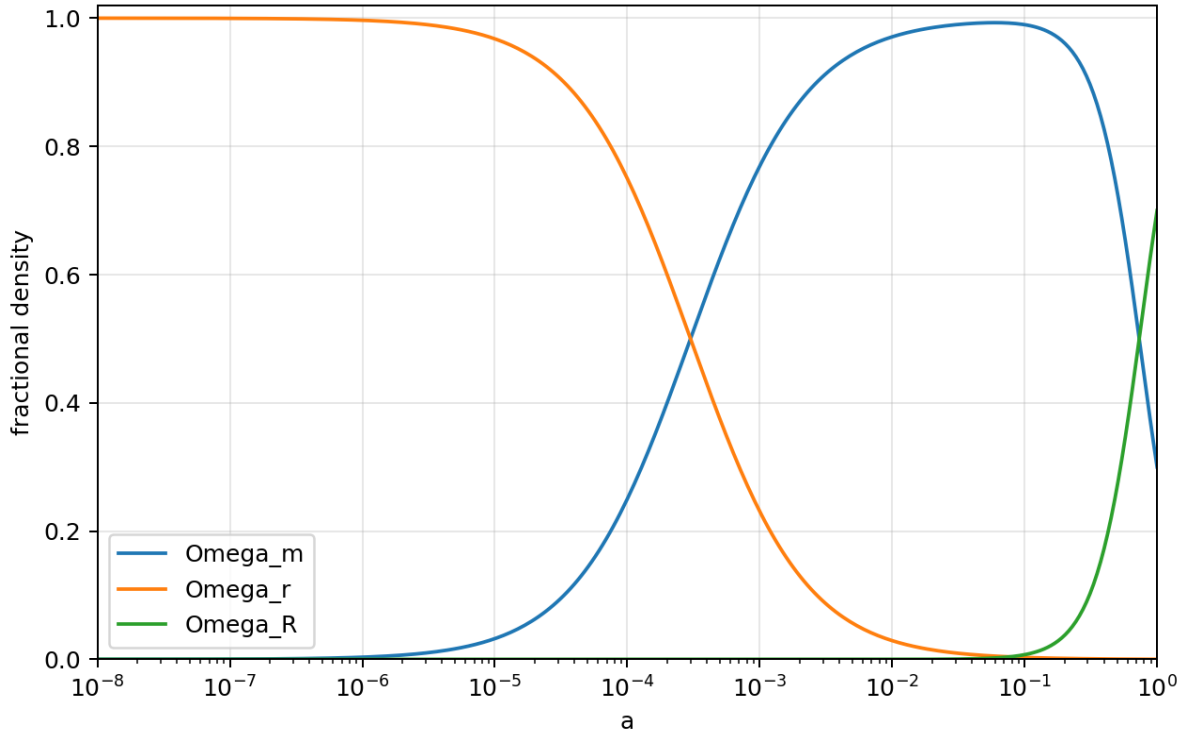


Figure 2. Fractional densities of the reference RFG-R branch. The regularization is absent at visible plotting precision on this cosmological domain.

4. Local GR matching and PPN

A single cosmological EFT is not automatically a local Solar-System EFT. RFG-R makes the crossover a model axiom. Define $W = \sqrt{|\text{abs}(C_{abcd} C^{abcd})|} / (H_0/c)^2$ and use a C-infinity switching function $s(W)$ that equals one for $W \leq 10^8$ and zero for $W \geq 10^9$. The complete low-energy action is $S_{\text{eff}} = S_{\text{GR}} + \int s(W)[L_{\text{cos}} - L_{\text{GR}}]$. FLRW has $W=0$, while a Schwarzschild Sun gives $W=5.756e22$ at one AU.

Inside the local matched domain $s=0$, so the action and all its variations are exactly those of GR. The PPN prediction is therefore $\gamma=1$, $\beta=1$, $\alpha_1=0$, and $\alpha_2=0$. The Cassini factor used by the likelihood is $-2 \ln L = [(\gamma-1-2.1e-5)/(2.3e-5)]^2$, yielding 0.833648 for the GR prediction.

5. Matter/CMB pre-Boltzmann gate and likelihood

No compressed CMB distance prior is used. The CMB interface begins with the action-level ADM derivatives. For $L = Q_{\text{epsilon}}(X)[R_3 + K_{ij} K^{ij} - K^2] + 2H_0^2 V_{\text{epsilon}}(X)$, the principal inputs include $L_{R_3} = Q_{\text{epsilon}}$ and $L_{K_{ij} R_3} = Q_{\text{epsilon},X} \gamma^{ij} / (3H_0)$. The exact ADM-to-extended-EFT map contains $\bar{m}_5 = -M_{\text{PI}}^2 Q_X / (3H_0)$, multiplying $\bar{m}_5 \delta R_3 \delta K/2$. The repository generates all background functions and derivatives in `eft_coefficients.csv`, avoiding any differentiation of a singular power law.

Pure-gravity scalar audit

The mapped $\bar{m}_5$ operator is retained in the reduced unitary-gauge scalar action. On a 49 by 49 logarithmic grid spanning $10^{-7} \leq a \leq 1$ and $10^{-4} \leq k/H_0 \leq 10^5$, the lapse-shift constraint discriminant remains positive but the zeta-dot-squared coefficient is degenerate at all 2,401 points; the maximum relative residual of its factorized numerator is $2.614e-16$. A standalone scalar sound speed is therefore undefined. This blocks a direct one-scalar EFT/Boltzmann evolution. It does not itself prove that the physical matter-coupled theory is ill-posed, since matter changes the constraint system.

Canonical-scalar check

The sourced scalar constraints have been reduced exactly for one minimally coupled canonical field in comoving gauge. If $m=\phi_{\text{dot}}^2/M_{\text{Pl}}^2$ and $D=-F_{XX}/3$, the lapse-shift determinant is proportional to $\Delta=6 D H^2 Q+(D-2Q)m$ and the reduced kinetic coefficient is $K=6 D Q m/\Delta$. The intrinsic-curvature contribution $+2Q(k^2/a^2)\zeta^2$ is retained; it is required to recover the exact GR massless-scalar result $K=6$ and $c_s^2=1$. On a self-consistent RFG-R reference branch with a massless canonical field, the executable diagnostic finds positive Δ , K , and gradient coefficients for $0.03\leq a\leq 1$. This is a real sourced-constraint check, not a dust/radiation/CMB calculation.

The required next derivation is the sourced photon-baryon-CDM-neutrino quadratic action from this same RFG-R action. Only after its full kinetic and gradient matrices, constraints, initial conditions, and GR limit are established may a 3+1 Einstein-Boltzmann backend evolve recombination and lensing. It must reject any point with a singular physical constraint matrix, a ghost, a gradient instability, or $Q_T\leq 0$. Accepted points would export TT, TE, EE, lensing, $P(k,z)$, $f\sigma_8(z)$, and the standard-siren distance.

6. Data protocol

The baseline log likelihood is $\ln L=\ln L_{\text{Planck18}}+\ln L_{\text{lensing}}+\ln L_{\text{BAO}}+\ln L_{\text{SN}}+\ln L_{\text{RSD}}+\ln L_{\text{PPN}}+\ln L_{\text{GW}}$. Planck is to be evaluated with the official low-ell temperature and polarization, high-ell TT/TE/EE, and lensing packages; the solver must be a full modified-gravity Einstein-Boltzmann solver rather than a quasi-static growth approximation. The public stock H-EFTCAMB interface is a useful GR reference, but it does not expose $\bar{m}_5$ δR_3 δK and is not an exact RFG-R solver. Cobaya may sample the joint likelihood only after the multi-fluid action reduction and extended solver are complete.

The completion defines a falsifiable route but has not yet supplied its decisive multi-fluid perturbation calculation. Whether the model is preferred, allowed, or ruled out remains an empirical question; no conclusion follows from background algebra, local matching, or the pure-gravity audit alone.

7. Numerical verification

check	reference result
high-X response recovery	5.329e-15 numerical maximum
high-X Q recovery	2.600e-15 numerical maximum
potential reconstruction residual	8.145e-06 finite-difference maximum
background density closure	1.554e-13 maximum error
minimum Q_T over a in $[1e-8,1e2]$	0.6153847956
canonical scalar GR limit	$K=6$, $c_s^2=1$ exactly
canonical reference grid	$\Delta>0$, $K>0$, $G_1>0$, $G_2\geq 0$ within numerical tolerance
pure-gravity extended-EFT scalar audit	2,401/2,401 degenerate; standalone c_s undefined
Solar W at 1 AU	5.756e22; local GR domain

The checks are generated by scripts/run_all.sh. They test the action-defined completion and reproduce its pure-gravity scalar gate; they do not constitute a CMB or matter likelihood.

References

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